

Modality-agnostic Fusion for Emotion Recognition

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Abstract

Modality-agnostic Fusion for Emotion Recognition is a challenging problem in human-computer interaction as the input may be in different, unknown modality. In this work, we focused on experimenting different modality-agnostic fusion mechanism in emotion recognition task, namely, early fusion, late fusion, and Gated Multi-modality Unit(GMU). We found that these fusion methods outperformed all uni-modal models on CREMA-D and MOSEI datasets. Moreover, we proposed a multi-task shared encoder for modality-agnostic fusion. The multi-task shared encoder is parameter efficient that can achieve a comparable performance to the fully fine-tuned models without training the original uni-modal encoders. Code is available at: <https://github.com/wenshicheng97/multimodal-emotion-recognition>

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1 Problem Definition

Emotion recognition is one of the most important tasks in the development of human-computer interaction, and it is one of the key technologies for machine understanding human communication. Previous works focused on single signals, typically audio, video, and text. Nevertheless, in multimedia, all these signals usually work in tandem to deliver the intended emotion of the person. Combining joint information from different resources to predict emotional states effectively would be a challenging task. As the complexity and variability of human emotions transcend the capacity of any single modality, the integration of multi-modal data through modality-agnostic fusion becomes crucial. Current multi-modality fusion strategies, including early fusion, late

fusion, etc., incorporate modality representation from different encoders to different layers of a fusion model. As these simple techniques last long in the multi-modal fusion realm, we aim to verify the performance of such fusion methods and develop a more complex fusion mechanism to enhance the performance of the multi-modal algorithms. Hence, we utilized a modified fusion mechanism, namely, a gated multi-modal unit that wisely handles the modality-agnostic fusion problem.

Moreover, these fusion mechanisms often require fully training a model or fine-tuning a pre-trained encoder, which may have a large memory in the training phase. Thus, we seek to find a shared and parameter-efficient architecture that can handle modality-agnostic representations and reach a comparable performance to the traditional fusion strategies that require fine-tuning uni-modal encoders. In this work, we proposed a shared encoder that projects different modalities input into low-rank representations using the same architecture. The shared encoder is parameter efficient and is able to handle the modality-agnostic fusion problem. The shared encoder achieved comparable performance to the basic fusion strategies on emotion recognition tasks in CREMA-D and MOSEI.

2 Literature review

Unimodal Encoders. Unimodal Encoders present a general methodology for learning representations of data in given modalities. Hsu et al. [2021] proposed a self-supervised speech representation learning approach, which utilizes an offline clustering step to provide aligned target labels for a BERT-like prediction loss, which learns representation from audio. Gong et al. [2021] introduced the Audio Spectrogram Transformer to learn audio representations by transforming audio into spectrograms and used ViT [Dosovitskiy et al. 2020] for audio classification. Kim and Lee [2023] proposed a novel method of improving ViT-based speech emotion recognition using coordinate information concatenate. For video encoders, Feichtenhofer et al. [2022] has shown that the masked autoencoder method can learn strong representations with almost no inductive bias on spacetime. Tong et al. [2022] and Cai et al. [2023a] proposed using masked autoencoders for self-supervised video representation learning, where Tong et al. [2022] masked video clips tube tokens in an extremely high ratio, and Cai et al. [2023a] masked tube tokens with facial region guidance.

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Multimodal Encoders. Similar to unimodal encoders, multi-modal encoders learn data representations from different modalities at the same time. Gong et al. [2022] proposed CAV-MAE by combining contrastive learning and masked data modeling to learn a joint and coordinated audio-visual representation. Yu et al. [2021] designed a label generation module based on a self-supervised learning strategy to acquire independent unimodal supervision and joint-trained the multi-modal and uni-modal tasks to learn the consistency and difference. Goncalves et al. [2023] proposed a shared conformer, VAVL, for audio and video input that can remain working when one of the modalities is missing at inference time.

Multimodal Fusion. Yin et al. [2022] proposed a fusion method by exchanging the normalized encoded hidden states from two modalities. Tsai et al. [2019] utilized cross-attention (replacing key and value with other modalities), and text can extract the information from audio and videos. Hu et al. [2022] proposed a multi-modal sentiment knowledge-sharing framework that unifies MSA and ERC tasks from features, labels, and models, which performed modality fusion at the syntactic and semantic levels and introduced contrastive learning between modalities and samples to better capture the difference and consistency between sentiments and emotions. Arevalo et al. [2017] proposed a gated multi-modal unit to fuse and find an intermediate representation of different modalities by learning to decide how modalities influence the unit's activation using multiplicative gates.

3 Data

3.1 Datasets

Table 1. Datasets statistics

label	CREMA-D(count)	MOSEI(count)
happiness	1271	4993
sadness	1271	888
anger	1271	637
disgust	1270	635
surprise	–	117
fear	1271	45
neutral	1087	–

3.1.1 CREMA-D. The Crowd-Sourced Emotional Multi-modal Actors Dataset[Cao et al. 2014] was designed for multi-modal emotion recognition tasks. It contains 7,442 samples acted by 91 actors, including 48 male and 43 female actors between the ages of 20 and 74 from a wide range of ethnic backgrounds. The actors spoke one of 12 sentences while expressing one of six different emotions: anger, disgust, fear, happiness, neutral, and sadness. Each emotional label has approximately 1,270 samples. Show in Fig. 1, the videos in

the dataset also have a relatively consistent duration and number of frames, ranging from about 40 to 140 frames per video.

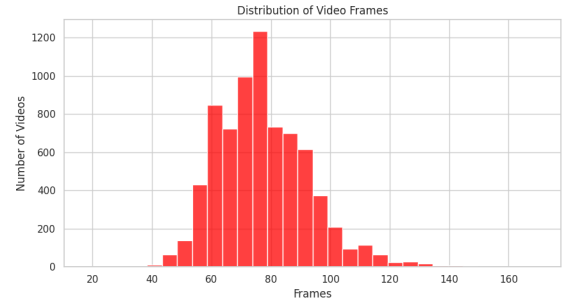


Figure 1. Distribution of video frame numbers in CREMA-D dataset

3.1.2 MOSEI. CMU Multimodal Opinion Sentiment and Emotion Intensity Dataset[Liang et al. 2023, 2021] was designed for multi-modal sentiment analysis and emotion recognition. It contains sentence utterance videos collected from more than 1000 YouTube speakers, and videos are randomly selected from various topics and monologue videos. Each video is annotated by human evaluator with six emotion labels: happiness, sadness, anger, disgust, surprise and fear. After filtering out some corrupted video files, we got 7315 usable data samples from the dataset. The video samples in the dataset is relatively long and have larger number of frames as Fig. 2.

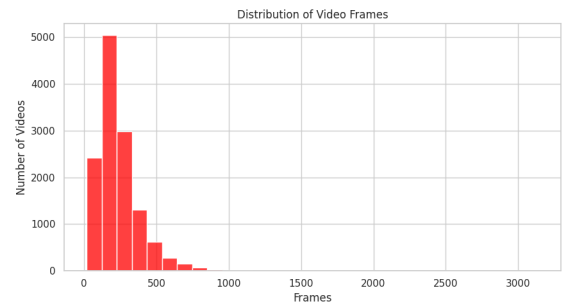


Figure 2. Distribution of video frame numbers in MOSEI dataset

3.2 Preprocess

3.2.1 Video.

OpenFace. OpenFace¹ [Baltrušaitis et al. 2016] is an open-source facial behavior analysis toolkit for facial behavior analysis. We used OpenFace to extract Action Unit 17 with intensity level and Action Unit 18 with presence value.

¹<https://github.com/TadasBaltrušaitis/OpenFace>

FaceX-Zoo. FaceX-Zoo [Wang et al. 2021] is an open-sourced tool for face recognition. We gave it original video samples as input, it returned face-cropped videos with dimensions of $224 * 224$ for each of samples.

MARLIN. We derived a facial representation from MARLIN (Masked Autoencoder for facial video Representation)[Cai et al. 2023b] by feeding in the face-cropped video generated by FaceX-Zoo. Due to CUDA memory limitations, it was not possible to process all frames from a video sample. We developed frame selection strategy for each dataset based on their video sample lengths. For the CREMA-D dataset, we selected 16 frames at a time with a temporal sampling rate of 2, capturing about half of a video content. Then since video sample in the MOSEI dataset contained generally larger number of frames, we evenly sampled 32 frames from each video sample.

3.2.2 Audio.

Spectrogram. We employed librosa [McFee et al. 2015], a Python package for audio analysis to get visual presentation of audio samples. We loaded audio samples with a sampling rate of 16 kHz and then constructed a mel-scaled spectrogram for each of them.

HuBERT. To utilize HuBERT [Hsu et al. 2021] for a better audio representation, we used the pre-trained Wav2Vec2 processor to extract audio features from wave files into tokens that can pass into the HuBERT model. We used the hubert-large-ls960-ft processor as the hubert-base does not come with a pre-trained processor.

3.2.3 TEXT. We utilized bert-base-cased model to tokenize each sentence in transcript samples.

3.2.4 CREMA-D. We created three custom datasets of CREMA-D for our experiment. The first dataset contained spectrogram as audio features and video action units 17 and 18 extracted by OpenFace as video features. We applied LSTM [Hochreiter and Schmidhuber 1997] model on this dataset as the baseline approach for our study. The second dataset contained audio features extracted by the pre-trained Wav2Vec2 and sampled video frames. This dataset was designed for Hubert and Marlin models with finetuning. The third dataset contained the same extracted audio features as the second dataset but incorporated facial representations from MARLIN model as video features. We applied frozen MARLIN model and linear probing techniques on the third dataset.

3.2.5 MOSEI. Video samples in MOSEI were collected from YouTube, and each video was annotated by three people. For each video sample, evaluators assign scores ranging from 0 to 3 for six labels: happiness, sadness, anger, disgust, surprise and fear. To obtain the combined label for experiment, we took the average label values if more than one

person scored that label above 0 and then applied majority vote across these labels. In case of a tie, we annotated the sample as neutral and excluded the sample from experiment since there is no Neutral label in the original dataset.

To align samples at the sentence level, we segmented the original video samples in dataset based on transcript segmentation. However, different segment videos may contain the same faces, resulting in data leakage. To deal with the problem, we assign segments from the same raw video sample exclusively to either train or validation dataset. Show in Table 1, there is a severe imbalance in the dataset, we applied a weighted sample strategy for the train dataset, selecting data samples based on the inverse ratio of their emotional label percentages.

4 Method

4.1 Uni-modality Baseline

We started with a simple LSTM classifier model as the baseline. For video data, we utilized features of Action Unit 17 (the value shows the intensity level) or Action Unit 18 (the value represents the presence) extracted by OpenFace [Baltrušaitis et al. 2016]. For audio data, we used mel spectrogram as features extracted by utilizing librosa.

4.2 Fusion

4.2.1 Early Fusion. We set HuBERT, MARLIN, and BERT (only for MOSEI dataset) as our base models for early fusion. Feature embedding generated by each model would first processed through a projector. Then the concatenation of projected features were fed into a classifier for further training. Additionally, each base model underwent finetuning to optimize their performances.

4.2.2 Late Fusion. Late fusion is only for model inference. We take the best unimodel trained previously. Base models are frozen and we sum up their logits after softmax as shown in equation 1. The summed logits will make the final decision.

$$O = \frac{1}{N} \sum_i Softmax(o_i) \quad (1)$$

4.2.3 Gated Multi-modal Unit. Inspired by Arevalo et al. [2017], we used a gated multi-modal unit for multi-modal fusion. For the k th modality's input x_k , the gate intermediate representation is computed as:

$$\begin{aligned} p_k &= W_{hk} \cdot x_k \\ h_k &= \tanh(p_k) \\ z_k &= \sigma(W_{sk} \cdot [x_1, \dots, x_k]) \\ g_k &= z_k * h_k \\ r &= \sum_{k=1}^N g_k \end{aligned}$$

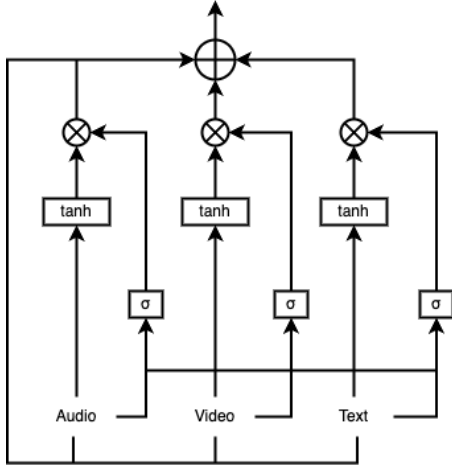


Figure 3. The architecture of the gated multi-modal unit. All inputs are projected to the same dimension before passing into the gate. σ is the learnable gate. $\otimes$ is the element-wise product. $\oplus$ is the addition operation. There are skip connections between the projected input and the representation after the learnable gate.

To avoid gates producing all zeros acrossing all modalities, we added a skip connection for the final intermediate representation

$$h = \text{Norm}(r + \frac{1}{N} \sum_{k=1}^N p_k)$$

A linear classifier will be added on the top of the GMU. The architecture of the unit is shown in Fig. 3.

4.2.4 Multi-task Shared Encoder. Inspired by [Goncalves et al. \[2023\]](#), we proposed a simplified shared encoder for multi-modal classification. The model will take pre-trained encoders for different modalities at first, these encoders remains frozen in training steps. Each encoder will produce a single representation vector for each sample. For audio and video inputs, we mean pooled the output representations of HuBERT and MARLIN, and for text input, we used the representation of the [CLS] token. Each representation vector will then pass through a projector to project to the same dimension. Then all projected vectors will go through a shared low rank encoder to learn a shared representation. Each shared representation will then pass through a modality specific decoder to learn a reconstruction of the original representation. The loss of unimodal part is defined as

$$\mathcal{L} = \mathbb{E} [\mathcal{L}_{\text{CE}}(\text{pred}, \text{real}) + \alpha \mathcal{L}_{\text{MSE}}(\text{rec}, \hat{r})] \quad (2)$$

Where rec is the reconstructed representation of an unimodality input, and $\hat{r}$ represents the corresponding original representation of the unimodal encoder. After each unimodality representation is backpropagated in each training step, we will freeze the projectors and the shared encoder, and

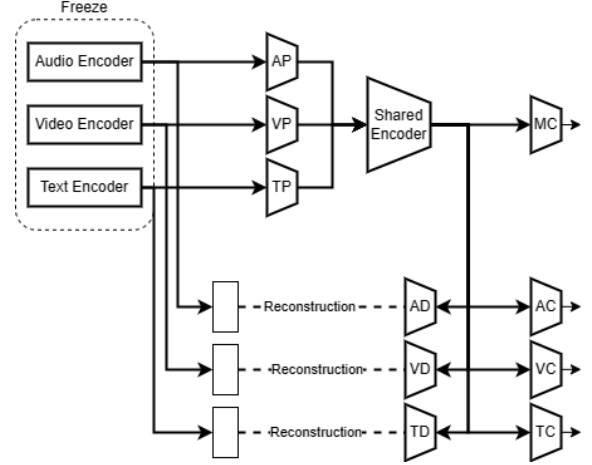


Figure 4. The architecture of the shared encoder. The pre-trained audio encoder, video encoder, and text encoder are frozen during training phase. AP, VP, TP are audio projector, video projector, and text projector. AD, VD, TD are audio decoder, video decoder, and text decoder. AC, VC, TC are unimodal audio classifier, unimodal video classifier, and unimodal text classifier. And MC is the multi-modal classifier.

Algorithm 1 Shared Encoder

Require: $D = \{A, V, T, \text{Label}\}$

Require: $\mathcal{M} = \{\theta_{ap}, \theta_{vp}, \theta_{tp}, \theta_s, \theta_{ad}, \theta_{vd}, \theta_{td}, \theta_{ac}, \theta_{vc}, \theta_{tc}, \theta_{mc}\}$

1: **procedure** TRAINING EPOCH

2: **for** $A_i, V_i, T_i, \text{Label}_i$ in D **do**

3: $\text{Pred}_a, \text{Rec}_a = M_{\{\theta_{ap}, \theta_s, \theta_{ad}, \theta_{ac}\}}(A_i)$

4: $\mathcal{L} = \mathcal{L}_{\text{CE}}(\text{Pred}_a, \text{Label}_i) + \alpha \mathcal{L}_{\text{MSE}}(\text{Rec}_a, A_i)$

5: backpropagate and update $\{\theta_{ap}, \theta_s, \theta_{ad}, \theta_{ac}\}$

6: $\text{Pred}_v, \text{Rec}_v = M_{\{\theta_{vp}, \theta_s, \theta_{vd}, \theta_{vc}\}}(V_i)$

7: $\mathcal{L} = \mathcal{L}_{\text{CE}}(\text{Pred}_v, \text{Label}_i) + \alpha \mathcal{L}_{\text{MSE}}(\text{Rec}_v, V_i)$

8: backpropagate and update $\{\theta_{vp}, \theta_s, \theta_{vd}, \theta_{vc}\}$

9: $\text{Pred}_t, \text{Rec}_t = M_{\{\theta_{tp}, \theta_s, \theta_{td}, \theta_{tc}\}}(T_i)$

10: $\mathcal{L} = \mathcal{L}_{\text{CE}}(\text{Pred}_t, \text{Label}_i) + \alpha \mathcal{L}_{\text{MSE}}(\text{Rec}_t, T_i)$

11: backpropagate and update $\{\theta_{tp}, \theta_s, \theta_{td}, \theta_{tc}\}$

12: Freeze $\{\theta_{ap}, \theta_{vp}, \theta_{tp}, \theta_s, \theta_{ad}, \theta_{vd}, \theta_{td}, \theta_{ac}, \theta_{vc}, \theta_{tc}\}$

13: $\text{Pred}_m = M_{\{\theta_{ap}, \theta_{vp}, \theta_{tp}, \theta_s, \theta_{mc}\}}(A_i, V_i, T_i)$

14: $\mathcal{L} = \mathcal{L}_{\text{CE}}(\text{Pred}_m, \text{Label}_i)$

15: backpropagate and update $\{\theta_{mc}\}$

16: **end for**

17: **end procedure**

18: **function** INFERENCE(A, V, T)

19: $\text{Pred} = \mathcal{M}(A, V, T)$

20: **return** Pred

21: **end function**

then the representation will be fused to learn a multi-modal

classifier for final classification. We use cross-entropy loss to learn the multi-modal classifier. The architecture of the model is shown in Fig. 4, and the functionality of the train and inference of the model is shown in Algorithm 1.

4.3 Experiment Setup

We utilized bert-base-cased for BERT, hubert-large for HuBERT, marlin-base for MARLIN. Parameter tuning was conducted through grid searches as shown in Table 2. We applied finetuning and linear probing for unimodality models, except for LSTM. BERT was only trained on MOSEI dataset since the CREMA-D dataset has the same sentence across all emotional labels. LSTM was trained on Action Units 17&18 and spectrogram as a baseline. Early fusion and gated fusion were fine-tuned on all base models, whereas late fusion only collected inference outputs from base models. And the shared encoder took frozen uni-modality representations for the reconstruction task. All experiments were conducted with fp16 precision due to the large memory requirement of processing video and audio data. Small batch sizes (2 and 4) were applied when processing video or audio data. Experiments were run by NVIDIA RTX A6000, A40, and A100, depending on availability. We utilized PyTorch Lightning, a more sealed deep learning framework, to run experiments and utilized Weights & Biases(wandb) for experiment logging. We then evaluated our model on accuracy and F-1 (micro) score using a validation dataset.

Table 2. Hyper-parameter set we have grid searched on

Hyper-parameters	Values
Learning rate	3e-4, 5e-4, 3e-5, 1e-3
Weight decay	1e-5
Batch size	2, 4, 16, 32
Projection size	256
Hidden size	512, 256, 128

5 Results

5.1 Unimodality Baseline

5.1.1 LSTM. The results are shown in Table 3 *LSTM* section. The first column represents what features are used. For CREMA-D dataset, action units 17 has the highest accuracy and F1 score, whereas spectrogram has the lowest performance. AU17 records the intensity of action units, which is more meaningful for model learning. AU18 only records the presence of action units, which could explain the slightly lower performance comparing to AU17. Spectrogram performs the worst with only around 0.5 accuracy and F1. We discussed that spectrogram has less meaningful information or the information is not significant for different emotions.

For MOSEI dataset, the performance is completely crashed. AU17 and AU18 are not enabling LSTM to learn at all since all

Table 3. Results on main experiments. Models are evaluated on CREMA-D and MOSEI validation dataset using accuracy and F1 (micro) score metric. Unimodal models have LSTM, fine-tuning and linear probing (freeze model) settings. Early fusion and gated fusion are fine-tuned on all base models. Late fusion takes sum of probabilities derived from base model inferences. Shared encoder only utilizes feature representations from base models without fine-tuning.

	CREMA-D		MOSEI	
	Acc.	F1	Acc.	F1
Unimodal Model				
<i>LSTM</i>				
AU17	0.730	0.730	0.432	0.432
AU18	0.647	0.647	0.432	0.432
Spectrogram	0.512	0.512	0.458	0.458
<i>Fine-tuning</i>				
BERT	–	–	0.689	0.690
HuBERT	0.798	0.79	0.692	0.690
MARLIN	0.814	0.814	0.656	0.658
<i>Linear Probing</i>				
BERT	–	–	0.69	0.69
HuBERT	0.70	0.70	0.639	0.639
MARLIN	0.732	0.733	0.632	0.632
Multi-modal Model				
Early Fusion	0.81	0.81	0.751	0.751
Gated Fusion	0.836	0.819	0.703	0.691
Late Fusion	0.880	0.880	0.702	0.702
Shared Encoder	0.774	0.758	0.685	0.613

training loss went to nan. Validation accuracy for these 2 features has not changed during the training step. We checked the experiment setting and models and ran several times but got same results, whereas LSTM with spectrogram can learn properly. We discussed that it might be CUDA version problem, or it could be that MOSEI dataset has longer video frames, resulting gradient exploding or gradient vanishing during training. LSTM can learn spectrogram properly but the result is not impressive. Compared to CREMA-D, MOSEI data is imbalanced and the spectrogram information is not significant to each emotion class, which could considerably influence the performance.

5.1.2 Fine-tuning & Linear Probing. Fine-tuning results refer to Table 3 *Fine-tuning* section. Models reach around 0.8 accuracy and F1 on CREMA-D and around 0.65 accuracy and F1 on MOSEI. Linear probing(freeze model) results refer to Table 3 *Linear Probing* section. Models reach around 0.7 accuracy and F1 on CREMA-D and around 0.63 accuracy and F1 on MOSEI. For these two settings, we could see HuBERT and MARLIN have comparable performance on emotion recognition. Only HuBERT on MOSEI dataset has slightly better

performance. It can be explained by that pre-trained models already handle detailed features representation—HuBERT for audio and MARLIN for face, perfectly suitable for downstream emotion recognition task. Regardless of solely taking the feature representation or fine tune the entire model, they all can perform well. Moreover, we recognize BERT in these 2 settings performs basically the same. We discussed that BERT representation is meaningful enough so there is no difference in these two settings. In fine-tuning setting, BERT is comparable to HuBERT and outperforms MARLIN, whereas BERT substantially outperforms them all in linear probing setting. Text always plays an important role in emotion and sentiment recognition.

5.2 Multi-modality

For multi-modality models, the late fusion mechanism achieved the highest performance on CREMA-D dataset, and the early fusion mechanism achieved the highest performance on MOSEI dataset. As we noticed the fine-tuned uni-modal models on CREMA-D dataset had already achieved a considerable performance, the confidence of the uni-modal models can highly affect the performance of late fusion model, bringing a way higher performance as the uni-modal models are trained individually. On the other hand, the participation of text features brought huge impact on the performance of multi-modal models on MOSEI dataset. As mentioned in Section 5.1.2, BERT outperformed other pre-trained uni-modal encoders, bringing significant improvement at fusion phase. Benefited from the superior performance of BERT, the early fusion, GMU, and late fusion mechanism outperformed all uni-modal models. The meaningful context representation may bring a higher confident score on BERT thus resulting a high performance in these fusion methods. Unfortunately, the multi-task shared encoder does not performed as well as other fusion strategies. However, it achieved a comparable performance to the fine-tuned uni-modal methods with less trainable parameters.

5.3 Parameter Efficiency

Table 4. Number of trainable parameters of models in fine-tuning settings on MOSEI dataset. M stands for million.

Model	Parameter
Early Fusion	510M
Gated Fusion	447M
MARLIN	86.4M
HuBERT	315M
BERT	108M
Shared Encoder	3.2M

In this section, we discuss parameter efficiency of our shared encoder architecture compared to other unimodality

and multimodality models. Number of trainable parameters is shown in 4. Early fusion and gated fusion have the most parameters since they incorporate all base models with the fine-tuning setting. For a unimodality model, HuBERT has the most parameters and MARLIN has the least parameters. In our shared encoder method, it only contains 3.2M parameters, significantly less than any models, because it only have a MLP encoder and several MLP decoders. Without massive trainable parameters, it still can reach a comparable performance to other fusion models.

6 Conclusion

In this work, we utilized HuBERT, MARLIN, and BERT models for emotion recognition on CREMA-D and MOSEI datasets. We performed uni-modality experiments and multi-modality experiments using 4 fusion mechanisms, namely, early fusion, late fusion, gated multi-modal and multi-task shared encoder. The early fusion, late fusion, and GMU outperformed all uni-modal models for emotion recognition task on CREMA-D and MOSEI. Multi-task shared encoder, with its low trainable parameter number nature, is not performing as well as other multi-modal methods, but still reached a comparable performance with other fine-tuning methods, showing its potential of being a parameter efficient multi-modal model.

7 Future Plan

Due to a lack of computational resources, we need to simplify the task of video pre-process. For future work, we aim to take more frames that may contain more information in face expression changes. We are also planing to bring more complex architecture in the multi-task shared encoder to improve its performance and remain the parameter efficiency nature.

8 Team members' contributions

Shicheng Wen. Literature review, built gated multi-modal unit and shared encoder architecture and preprocessed audio for HuBERT. Ran experiments on training HuBERT with and without fine-tuning, gated-fusion model, and shared encoder.

Yiming Tang. Preprocessed videos using OpenFace, pre-processed videos for MARLIN features. Built MARLIN model with fine-tuning and linear probing techniques, early fusion, late fusion. Ran experiments and evaluated all models on F1 score.

Ran Liu. Preprocessed audio data for LSTM training, pre-processed MOSEI dataset for training. Built LSTM model and BERT model. Ran experiments on training BERT and MARLIN.

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